

# SSL Certificate Viewer and Validator Tool using OpenSSL and GUI

## 1. Introduction

In today's internet-driven world, SSL/TLS certificates are essential for securing data transmission between clients and servers. An SSL certificate ensures that sensitive information such as passwords, credit card numbers, and other personal data remains encrypted during transmission. The goal of this project is to develop a Java desktop application that allows users to retrieve and validate SSL certificates from a domain using OpenSSL commands and a user-friendly GUI built in JavaFX or Swing.

This tool is particularly useful for developers, security analysts, and students who need to check certificate details and validate the authenticity of a website's SSL setup without using browser-based tools.

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## 2. Objectives

The primary objectives of this project are:

- To create a Java GUI application that accepts domain input (e.g., `www.example.com:443`)
  - To retrieve the SSL certificate using OpenSSL commands executed from Java
  - To parse and extract certificate details such as:
    - Subject
    - Issuer
    - Validity period
    - Serial number
  - To analyze the certificate's validity based on system date and provide real-time status feedback
  - To classify and display certificate status as "Valid", "Expired", or "Invalid Certificate"
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## 3. Tools and Technologies Used

| Tool/Technology      | Purpose                                         |
|----------------------|-------------------------------------------------|
| Java (JDK 8+)        | Core application logic                          |
| JavaFX / Swing       | GUI components                                  |
| OpenSSL              | Fetching SSL certificate from server            |
| ProcessBuilder       | Run shell/terminal commands in Java             |
| Date & Time APIs     | Validity check logic                            |
| Regex/String Parsing | Extract specific fields from certificate output |

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## 4. System Design and Working

The system architecture includes a front-end GUI and a back-end module to execute OpenSSL commands. The user enters a domain, and the application triggers OpenSSL to retrieve the SSL certificate. The returned text is parsed, and key fields are displayed. The tool compares the certificate's validity period with the system's current date to determine and display certificate status.

## Workflow:

1. User inputs domain (e.g., `www.google.com:443`)
2. The app uses `ProcessBuilder` to run `OpenSSL`
3. Certificate details are extracted from the result
4. Java compares the certificate's validity dates to the current date
5. GUI displays:
  - Subject
  - Issuer
  - Validity period
  - Serial number
  - Certificate status

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## 5. Implementation Highlights

- **OpenSSL Command Used:**
  - `openssl s_client -connect www.example.com:443 < /dev/null | openssl x509 -noout -text`
- **Parsing Strategy:**

Java reads OpenSSL output line by line and extracts values based on known keys such as `Subject:`, `Issuer:`, `Not Before:`, and `Not After:`.
- **Certificate Status Check:**
  - If the current date is before the `Not Before` date: “Invalid (Not Yet Valid)”
  - If it is after the `Not After` date: “Expired”
  - Else: “Valid”
- **GUI Interface:**
  - Input field for domain
  - Button to fetch certificate
  - Text area for displaying details
  - Label for certificate status

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## 6. Results

Example results for test domains:

| Domain                                  | Status  | Notes                                         |
|-----------------------------------------|---------|-----------------------------------------------|
| <code>www.google.com:443</code>         | Valid   | Certificate is within the validity period     |
| <code>expired.badssl.com:443</code>     | Expired | Certificate has passed its validity end date  |
| <code>self-signed.badssl.com:443</code> | Invalid | Certificate Not signed by a trusted authority |

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## 7. Conclusion

This project successfully demonstrates how Java can interact with external system tools like `OpenSSL` to create a useful GUI-based SSL certificate viewer. It helps users validate certificates for any accessible domain and ensures they are aware of expired or invalid security configurations. With further enhancements like PDF export or domain list batch checking, this tool could evolve into a practical asset for web administrators and cybersecurity students.



